

Curtis Fenner

TOKYO, JAPAN

curtisfenner.com/resume
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I am a highly experienced and deeply curious software engineer, specialized in careful knowledge of programming language and database semantics and performance.

I have extensive professional experience designing, building, and maintaining backend systems in statically-typed languages, as well as experience developing frontend applications with functional frameworks like React.

EXPERIENCE

Givery — Software Engineer, Track Test & Common Tokyo, Japan. Jan 2023 – current

- » Backend engineer on the Track Test product and on common systems
- » Track Test Team dev. lead (2025–): estimated, designed, and refined plans for long-term projects
- » Led architecture design, technology choice and modernization, and implementation of new product
- » Redesigned slow SQL queries to eliminate load and change slow-loading pages to instantly loading ones
- ⚙ **Scala (Play, cats), TypeScript (NestJS + Kysely) with PostgreSQL** (backends).
- ⚙ **TypeScript (React)** (frontend)
- ⚙ **AWS (ECS, Lambda, Rekognition, S3)**

PayPay — Software Engineer, Online ** Remote, Jun '21 – Apr '22 ** Tokyo, Japan. Apr 2022 – Jan 2023

- » Backend engineer on the Online team, which external APIs for accepting PayPay payments online
- » Designed, documented, and implemented new APIs used by merchants to accept PayPay payments
- » Designed and lead a new microservice, using requirements from Product Managers and other teams
- ⚙ **Kotlin & Java (Spring Boot) with MySQL** (backend)
- ⚙ **AWS (EC2, KMS)**

Square — Software Engineer, Orders API Atlanta, GA. Aug 2018 – Dec 2020

- » Backend engineer on the Orders API team, which operates both a public REST API and internal POS API
- » Refactored, tested, thoroughly documented, and simplified complex validation, transaction calculation, and public API versioning logic
- ⚙ **Java, MySQL, and Protobuf** (strongly consistent API server).
- ⚙ **TypeScript (React)** (internal web UI)

INTERNSHIPS

Qualtrics — Software Engineer Intern, Data Platform Seattle, WA. Summer 2017

- ⚙ **Scala** and **Elasticsearch** for customer report generation
- » Redesigned and reimplemented statistical weighting feature to correct formulas

Qumulo — Software Engineer Intern, Filesystem Performance Seattle, WA. Summer 2016

- ⚙ **C** (filesystem server) and **Python** (test automation and code generation)
- » Eliminated lock contention in a multithreaded cache to reduce file operation latency

Square — Software Engineer Intern, Public API San Francisco, CA. Summer 2015

- ⚙ **Node.JS** and **Go** (microservice backends)
- » Optimized and refactored public API server to halve average response time

EDUCATION & LANGUAGE

University of Michigan: Computer Science B.S.E, 4.0 GPA Ann Arbor, MI. 2014 – 2018

- » **Selected Coursework:** Distributed Systems (W2017), Grad. Programming Languages (F2017)
- » Teaching assistant for Distributed Systems using Golang (F2017)
- » World Finalist in 2017 ACM International Collegiate Programming Contest (ICPC)
- » Japanese Language I + II

Japanese Language Proficiency Test N3 (JLPT/日本語能力試験) Tokyo – July 2024

SELECTED PROJECTS

Tokyo Rail-Transit Isochrone github.com/CurtisFenner/tokyo-transit-iso

- » Implemented routing algorithm and region geometry rendering algorithm on real transit schedule data

Shiru Programming Language & Compiler github.com/CurtisFenner/shiru-ts

- » Designed and implemented a toy programming language, compiler, and assertion verifier, including a PEG parser, a type-checker, a simple SMT solver, and a JavaScript code generator.